

Code:

```
#include <stdio.h> #include <stdlib.h>

// Dynamic coordinate tracking using ordered set representation typedef struct Node
{ int val; int priority; struct Node *left, *right; } Node;

Node* new_node(int val) { Node* n = (Node*)malloc(sizeof(Node)); n->val = val; n-
>priority = rand(); n->left = n->right = NULL; return n; }

void split(Node* t, int key, Node** l, Node** r) { if (!t) { *l = *r = NULL; } else if (t->val <=
key) { split(t->right, key, &(t->right), r); *l = t; } else { split(t->left, key, l, &(t->left)); *r = t; } }

void insert(Node** t, Node* item) { if (!*t) { *t = item; } else if (item->priority > (*t)-
>priority) { split(*t, item->val, &(item->left), &(item->right)); *t = item; } else { insert(item-
>val < (*t)->val ? &((*t)->left) : &((*t)->right), item); } }

int lower_bound(Node* t, int key) { int ans = -1; while (t) { if (t->val >= key) { ans = t->val; t =
t->left; } else { t = t->right; } } return ans; }

int main() { int n, q; if (scanf("%d %d", &n, &q) != 2) return 0;

Node *root = NULL;
int i;
for (i = 0; i < q; i++) {
    int type, val;
    scanf("%d %d", &type, &val);
    if (type == 1) {
        insert(&root, new_node(val));
    } else {
        int res = lower_bound(root, val);
        printf("%d\n", res);
    }
}

return 0;

}
```

Input:

```
5 5
2 3
1 2
2 1
2 3
2 2
```

Output:

```
-1 2 -1 2
```